
Refinement Steps Buy Intelligibility, Not Identity: Metric-Selective Test-Time Scaling in Masked-Diffusion TTS

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Abstract

1 A diffusion language model buys sequential computation twice: depth, paid once
2 in parameters, and denoising steps, rented per utterance at serve time. We ask
3 whether the rented kind helps every capability equally. We train an iso-parameter
4 width–depth grid of 15 masked-diffusion codec TTS models (2 seeds, 20–125 M
5 non-embedding parameters) and sweep the refinement budget T at inference only,
6 giving a 75-point (w, d, T) surface per metric from 30 trained models. Fitting $\text{err} =$
7 $E + Aw^{-\alpha} + Bd^{-\beta} + CT^{-\tau}$ pre-registered, refinement steps attack intelligibility
8 harder than speaker identity: $\Delta\tau = \tau_{\text{WER}} - \tau_{\text{SIM}} = 0.065$, 95% run-level bootstrap
9 CI [0.053, 0.077]. Word error rate saturates only at $T^*=16$ while identity saturates
10 at 8. Depth and steps are *not* interchangeable: the substitution form $B(dT^\kappa)^{-\beta}$
11 is worse than the separable form by $\Delta\text{AICc} = +55.2$. Steps are an intelligibility
12 knob, not a general one.

1 Introduction

14 A diffusion language model can buy sequential computation twice. It *owns* depth: d transformer
15 blocks, paid once in parameters and then on every forward pass forever. It *rents* refinement steps:
16 T denoising passes over the same sequence, paid per utterance and adjustable at serve time. An
17 autoregressive model has only the first lever; its serial cost per token is fixed by the architecture the
18 moment training ends. If rented serial compute substitutes for owned serial compute, a deployment
19 can trade parameters for steps after the fact — but only for the capabilities that steps actually improve.

20 Zero-shot text-to-speech is an unusually clean probe of that question, because it separates two
21 capabilities that are measured by standard, independent metrics. Intelligibility is compositional: the
22 model must place the right phonetic content in the right order, and word error rate (WER) reads it
23 out. Speaker identity is representational: the model must carry timbre from a short prompt into new
24 content, and speaker similarity (SIM-o) reads that out. The two need not respond to the same knob.

25 We train one iso-parameter width–depth grid of masked-diffusion codec language models and then
26 sweep T at inference only, which yields a full (w, d, T) response surface per metric at zero additional
27 training cost. Two claims were pre-registered before any run. **H-D2**: refinement steps improve
28 intelligibility more than identity, i.e. test-time scaling is metric-selective. **H-D3**: for intelligibility,
29 steps and depth exchange at a measurable rate κ — how many step-doublings buy one depth-doubling.

$$\text{err}_m(w, d, T) = E_m + A_m w^{-\alpha_m} + B_m d^{-\beta_m} + C_m T^{-\tau_m} \quad \text{err}_m(w, d, T) = E_m + A_m w^{-\alpha_m} + B_m (dT^{\kappa_m})^{-\beta_m} \quad (1)$$

Both claims were registered before the first run, together with the thresholds, the frozen sampler and the analysis plan. The answer to the first is yes and quantitative: $\Delta\tau = 0.065$ with a run-level bootstrap CI of $[0.053, 0.077]$, so the step exponent is reliably larger for intelligibility than for identity. The answer to the second is a clean no: the substitution form is decisively *worse* than the separable one ($\Delta\text{AICc} = +55.2$ for WER, $+89.8$ for SIM-o) and κ runs to its bound, so no finite depth-step exchange rate describes this regime. The two results together say something sharper than either alone: extra inference compute is a targeted instrument, and it cannot be treated as a substitute for architecture.

Contributions. (i) The first per-metric (w, d, T) scaling surface for speech generation, 75 points per metric from one training grid, released with the harness. (ii) $\Delta\tau$ with a confidence interval, and the negative result on depth-step substitution stated as an AICc comparison rather than a narrative. (iii) A serving consequence measured rather than assumed: on the evaluation GPU c_{layer} is flat in width at batch 1, so serial latency is $8Td \cdot c_{\text{layer}}$ and the step budget is the only latency knob that trades against intelligibility.

2 Related work

Masked diffusion language models are now trained at scale [Nie et al., 2025, Sahoo et al., 2024, Austin et al., 2021], but their scaling is reported against a single likelihood-shaped objective, so a per-capability question cannot arise. In speech, the masked-generative recipes that made codec-token TTS practical — confidence-decoded parallel generation over residual codebook levels [Borsos et al., 2023, Chang et al., 2022] and its multi-stage successors [Wang et al., 2024] — fixed a step budget by engineering judgement and reported a single operating point; none maps error against steps as a law. Autoregressive codec TTS [Wang et al., 2023] has begun to report scaling behaviour, but in total parameters N only [Ye et al., 2025], which cannot separate width from depth and has no step axis at all; classical scaling work is likewise N -indexed [Kaplan et al., 2020, Hoffmann et al., 2022]. Test-time scaling is well studied in reasoning [Snell et al., 2024], where the payoff is measured on one correctness metric; whether extra inference compute helps different capabilities *differently* is, to our knowledge, untested in generative speech. That is the gap this abstract fills: per-metric laws in (w, d, T) for a masked-diffusion TTS model.

3 Method

Backbone and objective. A bidirectional (non-causal) pre-LN transformer over Mimi [Défossez et al., 2024] codec tokens at 12.5 Hz with the first 8 residual codebooks (2048 entries each, plus MASK and PAD). Width w , depth d , head dimension 64, GELU MLP ratio 4, RMSNorm, RoPE, no weight tying, so the non-embedding parameter count is $N = 12dw^2$. One embedding table per codebook is summed at the input and one linear head per codebook reads out. Training is SoundStorm-style [Borsos et al., 2023] coarse-to-fine masking: per example a level $l \sim \mathcal{U}\{1..8\}$ and a ratio $t \sim \mathcal{U}(0, 1]$; levels below l stay visible, level l is masked i.i.d. with probability t , levels above l are entirely MASK; the loss is cross-entropy on the masked cells of level l only. A phoneme prefix and a 3-second prompt of codec tokens are always visible; there is no speaker encoder, so the backbone itself is the only identity pathway, and no classifier-free guidance, which would add a second inference-time axis and confound the step axis. The target length is fixed before generation from the text length and a corpus character rate (0.0602 s/char).

Frozen sampler. MaskGIT-style confidence decoding [Chang et al., 2022], level by level, with exactly T refinement steps per level, so $\text{NFE} = 8T$: a cosine schedule sets how many cells remain masked after each step, tokens are drawn categorically at temperature 1.0, and confidences are log-probabilities plus Gumbel noise annealed $1 \rightarrow 0$ within a level; committed cells stay committed. The sampler is identical for every config, seed and T — nothing is tuned per shape or per step budget. Per-item generation noise is keyed so that each item sees the same noise at the first step of every level for every T , making T the only difference between surface points of one run. Before any metric is computed we verify that T reaches the sampler: outputs at $T=1$ and $T=16$ must differ on more than 20% of generated token cells.

Design and statistics. 15 configs span three iso- N budgets $\times$ five aspect ratios, each trained at 2 seeds for 30,000 steps at effective batch 256 with μP [Yang et al., 2021] (base width 256, base LR 0.004); width transfer and depth transfer were verified by five-point LR sweeps before the grid. $T \in \{1, 2, 4, 8, 16\}$ is swept at evaluation only, giving 75 surface points per metric from 30 trained models. Metrics: WER (Whisper-large-v3 [Radford et al., 2023], greedy, the Whisper English normaliser), SIM-o (WavLM-large [Chen et al., 2022] speaker-verification embedding against the *original* prompt waveform, following Anastassiou et al. [2024]), a frozen degenerate-output rule, and UTMOS [Saeki et al., 2022] as a tertiary. Error transforms are $\text{err}_{\text{WER}} = \text{WER}$ and $\text{err}_{\text{SIM}} = 1 - \text{SIM-o}$, computed over all items including degenerates. Both forms of Eq. (1) are fitted by weighted non-linear least squares with 32 multi-starts; $\Delta\tau = \tau_{\text{WER}} - \tau_{\text{SIM}}$, κ , and $\Delta\rho = \alpha_{\text{SIM}}/\beta_{\text{SIM}} - \alpha_{\text{WER}}/\beta_{\text{WER}}$ carry percentile confidence intervals from a bootstrap that resamples at the *run* level — the five T -points of one run share a trained model, so resampling surface points would understate their correlation — with 2000 replicates recomputing every T point.

4 Results

Steps are metric-selective (H-D2, supported). Fitted on all 75 surface points, the step exponent is $\tau_{\text{WER}} = 0.809$ (CI [0.803, 0.814]) against $\tau_{\text{SIM}} = 0.744$ (CI [0.732, 0.755]), so $\Delta\tau = 0.065$ with CI [0.053, 0.077], excluding zero. The raw surface shows the same thing without any model: averaged over the grid, going from $T=1$ to $T=16$ takes WER from 113.1% to 17.7% — a factor of $6.4\times$ on the error scale — while SIM-o moves only from 0.208 to 0.371, a factor of $1.26\times$ on $1 - \text{SIM-o}$. The saturation steps differ accordingly: $T_{\text{WER}}^* = 16$, i.e. intelligibility is still improving at the largest budget we tested, while $T_{\text{SIM}}^* = 8$. The power gate over shapes passed in all three budgets before these tests were run.

Depth and steps do not substitute (H-D3, not supported). The pre-registered substitution form collapses: $\Delta\text{AICc}(M_{\text{sub}} - M_{\text{sep}}) = +55.2$ for WER and $+89.8$ for SIM-o, where the hypothesis required $\leq +4$, and the fitted κ sits on its bound for both metrics, i.e. the exchange rate is not identified. We therefore quote no κ . The mechanism is visible in the fits: T enters with a large additive term that decays as $T^{-\tau}$ towards a floor, whereas depth changes the floor itself; a model that forces the two into one factor dT^κ cannot fit both the strong depth ordering at fixed T and the strong T ordering at fixed depth.

Shape matters, but the anisotropy ratio is not identified (H-D1, not supported). At $T=16$ the two-axis model beats the N -only model decisively for both metrics ($\Delta\text{AICc} = -29.5$ for WER, -16.3 for SIM-o, against a -4 threshold), so error is not a function of parameter count alone — iso- N shape choice matters. The direction of the anisotropy, however, we cannot claim: $\Delta\rho = -0.615$ with CI $[-1.005, 1.011]$, spanning zero, and the width coefficient A collapses to its lower bound for both metrics at $T=16$, which leaves α — and hence $\rho = \alpha/\beta$ — unidentified. What the data do show unambiguously is a strong depth ordering at fixed parameter count: within each budget, the deepest shape beats the shallowest by 33.5 (20 M), 16.5 (50 M) and 5.6 (125 M) points of WER, while SIM-o varies by less than 0.065 across all 15 shapes.

Extrapolation and reliability. Fitting the two smaller budgets and predicting the largest gives MAPE 2.2% for SIM-o (versus 3.0% for the N -only model), inside the 15% the hypothesis asked for; WER does not extrapolate (24.7% versus 23.2%). Degenerate outputs — empty, truncated or looping transcripts under the frozen rule — fall monotonically with the step budget, from 41.0% at $T=1$ to 0.1% at $T=16$: most of what $T=1$ costs is not subtle quality but outright failures.

Serving corollary. Measured on the evaluation GPU at batch 1, c_{layer} is flat in width across the whole grid (0.493–0.514 ms per layer for $w \in [256, 1152]$): at these sizes the device is launch-bound, so serial latency is $8Td_{\text{c}_{\text{layer}}}$ and width is free. Combined with the two results above, the allocation rule is unusually simple. Spend parameters on depth until the latency budget binds, because depth lowers the WER floor and width does not measurably move either metric’s floor here; then spend the remaining latency on steps, which buy intelligibility down to $T_{\text{WER}}^* = 16$ but stop buying identity after $T_{\text{SIM}}^* = 8$. A system that needs identity more than intelligibility — voice preservation with short, fixed prompts — can run at 8 steps and reclaim 50% of the serial latency of a $T=16$ configuration.

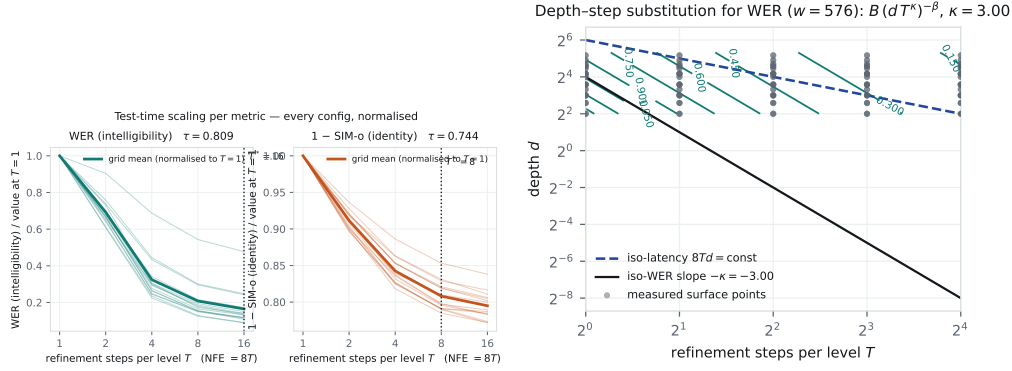


Figure 1: **Left:** error versus refinement steps for every config, normalised to its own $T=1$ value, with the pooled saturation step T^* marked; WER (teal) keeps falling to $T=16$ while SIM-o (orange) flattens by $T=8$. **Right:** iso-WER contours from M_{sub} in $(\log T, \log d)$ with the iso-latency line $8Td = \text{const}$; the fitted κ is at its bound, which is why the substitution reading of these contours is rejected in the text ($\Delta\text{AICc} = +55.2$).

Table 1: Fitted exponents on the 75-point surface (M_{sep} for α, β, τ ; M_{sub} for κ) with 95% run-level bootstrap CIs for the step exponents. $\Delta\tau = 0.065$, CI [0.053, 0.077], excludes zero: the pre-registered primary result. κ is at its bound in both rows and $\Delta\text{AICc}_{\text{sub-sep}} \gg +4$, so no exchange rate is quotable. $\Delta\text{AICc}_{\text{full-N}} \leq -4$ means iso- N shape matters.

Metric	α (width)	β (depth)	τ (steps)	κ	$\Delta\text{AICc}_{\text{sub-sep}}$	$\Delta\text{AICc}_{\text{full-N}}$
WER	2.951	1.912	0.809 [0.803, 0.814]	3.000 [†]	+55.2	-29.5
SIM-o	0.917	0.672	0.744 [0.732, 0.755]	3.000 [†]	+89.8	-16.3

[†] at the fitted bound, i.e. not identified; $\Delta\tau = 0.065$, CI [0.053, 0.077].

5 Limitations and outlook

The ceiling is 133 M non-embedding parameters and 30,000 steps; every model is under-trained by frontier standards, and the T curves may flatten differently at scale — $T_{\text{WER}}^* = 16$ is a lower bound, since WER had not saturated at the largest budget we measured. One language, one codec, one corpus, and no classifier-free guidance: guidance is a second inference-time axis and would have confounded the step axis, so excluding it is deliberate but it does leave the most obvious follow-up open. The $\Delta\tau$ interval comes from resampling 2 runs per config, so it captures seed-to-seed variation and not the uncertainty contributed by the single shared harness and corpus; it should be read as the pre-registered statistic it is, not as a full error budget. $\Delta\rho$ and κ are reported as unidentified rather than as estimates, and no claim is made from them. Calendar and compute cuts taken: none. Under the pre-registered outcome classes this is class S1, and the selectivity claim is the only positive claim the data license.

Reproducibility. The protocol, hypotheses, thresholds, frozen sampler and analysis plan were registered before the first training run, and every gate evaluation is logged with its measured values. The released artefact contains the harness, all 30 run records, the full 75-point surface, the fits and the bootstrap code. Data is a 2000 h English subset of a public corpus [He et al., 2024] with 64,680 training speakers and 200 speakers held out; the eval set is curated only for ground-truth transcribability, a model-independent filter. Total cost 143 GPU-hours on one 8-GPU node.

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